

Anomaly Detection

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Introduction

Anomaly detection flags observations that deviate from typical patterns. It is used for fraud detection, sensor fault monitoring, network intrusion, and adverse-event surveillance. Unlike classification, labels for anomalies are often unavailable at training time – methods are unsupervised or semi-supervised.

Prerequisites

Multivariate data; distance / density concepts.

Theory

Distance-based: kNN distance or average distance to neighbours; high = anomaly.

Density-based: LOF (Local Outlier Factor) compares local density to neighbours.

One-class SVM: learns a boundary enclosing most of the training data; points outside are anomalies.

Isolation Forest: exploits that anomalies are easier to isolate in random trees (shorter path length to terminal node).

Autoencoders: train a reconstruction network on normal data; anomalies have high reconstruction error.

Assumptions

Training data is mostly “normal”; anomaly pattern differs in some quantifiable way from normal.

R Implementation

```
library(e1071)

set.seed(2026)
n <- 300
# Normal data is a ball; add some outliers
df <- data.frame(
  x1 = c(rnorm(n, 0, 1), runif(20, -5, 5)),
  x2 = c(rnorm(n, 0, 1), runif(20, -5, 5))
)
labels <- c(rep("normal", n), rep("anomaly", 20))

# One-class SVM
oc <- svm(df, type = "one-classification", nu = 0.1, kernel = "radial")
pred <- predict(oc, df) # TRUE = inlier, FALSE = anomaly

table(truth = labels, flagged = ifelse(pred, "inlier", "anomaly"))
```

Output & Results

Confusion matrix of the one-class SVM against ground truth; it correctly flags most of the 20 injected anomalies as outliers.

Interpretation

“One-class SVM detected 17 of 20 injected anomalies (sensitivity 0.85) with a 6 % false-positive rate on normal data. Isolation Forest gave similar results with less tuning.”

Practical Tips

- Start with Isolation Forest; it is fast, robust, and has few hyperparameters.
- `nu` in one-class SVM approximates the expected anomaly fraction.
- Autoencoders are competitive for complex high-dimensional anomalies (images, signals).
- Validate with a held-out labelled anomaly set; unsupervised evaluation is hard.
- Combine multiple detectors for robustness; anomalies are rare and single-method false-positive rates compound.

For Reviewers

What to look for in a paper using this method.

- Common misapplications.
- Diagnostics that should be reported but often aren't.
- Red flags in tables and figures.
- What to verify.
- What an adequate Methods paragraph must contain.

See also — labs in this chapter

- Cross-validation, nested CV, bootstrap .632+
- Regularisation: ridge, lasso, elastic net
- Trees, random forests, gradient boosting
- Interpretability and SHAP
- Tabular neural networks with torch
- Imaging and sequence models intro
- tidymodels pipelines
- FDR, knockoffs, replication crisis